

1. Title Page

Project title: Sleep Stage Classification Using Deep Learning

Course name: BIO3609 – Biomedical Signal Processing

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2. Abstract

Getting enough quality sleep is important for humans to function properly. To properly diagnose sleep disorders, sleep stage classification is an important process. Traditional methods are manually done by sleep experts using polysomnography signals, which makes them time-consuming and prone to human error. Automated systems are now being developed to help automate this process and give more accurate results using physiological signals. This report gives an overview of deep learning approaches for automatic sleep stage classification based on the electrocardiography signal. Recent research studies are also reviewed and show results using evaluation metrics such as accuracy and Cohen's Kappa score. Furthermore, the challenges of sleep stage classification, including data imbalance, overfitting, and difficulty in detecting REM and deep sleep stages, are highlighted.

3. Introduction

Sleep is body process that allows the body and brain to rest. People view it as simple as just closing your eyes and drifting off, but it is actually a complex body process. The number of sleep hours needed every night varies from one person to another and can change over time as well.

The American Academy of Sleep Medicine (AASM) manual offers a systematic approach to sleep scoring. It analyzes polysomnographic (PSG) recordings to classify sleep stages and events. The stages are as follows: wakefulness, non-rapid eye movement (NREM), and rapid eye movement (REM). NREM consists of three stages: N1 and N2 for light sleep, and N3 for deep sleep. This classification is usually made manually by sleep experts using the PSG recordings, which is usually time-consuming and subjective. It is necessary to correctly identify these stages to diagnose sleep disorders, monitor sleep quality, and understand body functions during sleep.

With the recent advancements in deep learning and artificial intelligence, powerful tools for automatic sleep stage classifications have emerged. Deep learning models are able to learn patterns directly from the physiological signals. The models can capture characteristics of sleep data, enabling more reliable detection of sleep stages.

This report will cover a from-scratch model that we coded to classify the sleep stages. It will discuss the pipeline used to build the model, the challenges faced, and the results achieved.

4. Problem Statement

Normal sleep is divided into different stages. To diagnose sleep disorders such as insomnia, sleep apnea, and narcolepsy, it is very important to accurately classify these sleep stages. Traditionally, the stages are classified manually by sleep experts using PSG recordings, which is time-consuming, labor-intensive, and subjective due to the expert's interpretation. To meet the demand for accurate and efficient sleep monitoring systems, reliable automated methods are needed.

Conventional machine learning techniques require manual feature extraction, which requires expert knowledge and may not succeed at capturing the complex patterns in physiological signals. Other than that, sleep data are nonlinear, unbalanced, and full of noise, which makes it more difficult to accurately classify, especially for deep sleep and REM. In this report, a deep learning model uses the ECG signal from the PSG recordings to try to classify the sleep stages according to the AASM standards.

The reason behind using the ECG signals to classify sleep stages is that they are easier to acquire, non-invasive, and could be adapted to wearable or home-based healthcare systems. Heart rate variability (HRV) extracted from ECG signals has shown a strong correlation between sleep stage transitions and sleep arousal events.

5. Literature Review

The research paper used in this project as a reference is 'A Multi-task Deep Learning Algorithm for Sleep Stage Scoring and Sleep Arousal Detection.' It proposes a deep learning framework to classify sleep stages and detect sleep arousal using a single-lead ECG signal. The model shares information between both tasks instead of training separate models independently.

The authors used the Multi-Ethnic Study of Atherosclerosis (MESA) dataset. After preprocessing and exclusion of incomplete records, it contained PSG records from 1,069 subjects. As primary input, ECG signals were sampled at 256 Hz. The four classes used were wake, light sleep (combined N1 and N2), deep sleep (N3), and REM sleep. A binary classification approach was used for arousal detection.

The architecture was based on the DeepCAD framework and included convolutional layers, residual blocks, batch normalization, Long Short-Term Memory (LSTM) units, and fully connected layers. The model employed a task-incremental multi-task learning strategy, where the network first learned arousal detection and then expanded to sleep stage

scoring while reusing learned features. This shared representation approach improved learning efficiency and reduced overfitting.

The results demonstrated strong performance for both tasks. The model achieved an AUROC of 0.94 for arousal detection and an overall sleep stage classification accuracy of 79% with a Cohen's Kappa score of 0.68. The model showed strong performance in classifying wake, light sleep, and REM stages, but deep sleep classification remained more challenging due to fewer deep sleep samples in the dataset.

The study also compared its results with previous deep learning methods. Earlier studies using ECG and HRV features with CNN and LSTM models achieved classification accuracies between 76% and 80%. For example, Sridhar et al. used instantaneous heart rate with a CNN model and achieved 80% accuracy with a Cohen's Kappa of 0.69. The multi-task model achieved comparable performance while also performing two related tasks using only a single-lead ECG signal.

Even though the study achieved great progress, it acknowledged several limitations. The model was trained and tested only on the MESA dataset, limiting generalizability to other datasets and deep sleep prediction accuracy remained relatively low. The researchers suggested that future work should evaluate the model on larger and more diverse datasets and explore improved feature extraction strategies.

6. Dataset Description

The MESA (Multi-Ethnic Study of Atherosclerosis) Sleep dataset is a publicly available dataset, collected from 2010 to 2013, designed to investigate the relationship between objectively measured sleep characteristics and cardiovascular disease in a diverse population of older adults. They had around 2,261 adults, ages 45-84. The diversity includes Non-Hispanic Whites, African Americans, Hispanic Americans, and Chinese Americans. The study has a nearly even gender split of ~54% females and ~46% males, making it qualified to investigate gender differences as well.

The dataset uses three modalities of sleep assessments, consisting of full-night PSG recordings, wrist actigraphy for 7 consecutive days, and sleep questionnaires. The PSG is a collection of physiological data that comprises multiple channels that collect EEG, EOG, EMG, and ECG signals, breathing patterns, oxygen levels, and leg movements. The wrist actigraphy is worn for 7 consecutive days on the non-dominant wrist to track sleep duration, wakefulness, and night-to-night sleep variability. The dataset is divided into signals (.edf) and annotations (.xml).

For this project, we are using the ECG signal to classify the sleep stages at a sampling frequency of 256Hz. Arousal is binary coded (0=no arousal, 1=arousal). The classes used are as follows: wake, light sleep (N1+N2), deep sleep (N3), and REM. Ten subjects were used to train, validate, and test.

7. Project Pipeline:

This project aims to answer the following question: “Are we able to classify a signal into the 4 sleeping stages (Wake, Light, Deep , REM) using only the ECG signal?”. The heart rate or ECG signal changes during sleep, depending on the sleeping stage. This is why both HRV features and the raw ECG waveform were used for sleep stage classification.

1. Dataset Access & Loading (MESA Sleep Study)
2. ECG Signal Extraction
3. Signal Preprocessing: filtering and Z-score normalization
4. Signal Segmentation (30-second epochs)
5. Multi-Scale Feature Extraction: Extraction of raw waveforms and Context-Aware HRV features (5.5-minute windows).
6. Data Balancing: Implementation of WeightedRandomSampler
7. Hybrid CNN-BiLSTM with Attention Mechanism
8. Advanced Optimization: Focal Loss and Cosine Annealing Learning Rate scheduling.
9. Model Training
10. Performance Evaluation

First, the dataset was loaded and each EDF file was matched with its corresponding XML annotation file . The sleep stage annotations were then mapped into numerical labels to make them suitable for supervised learning. The dataset contained only 10 subjects, which is considered a major limitation for deep learning training.

```
STAGE_MAP = {
    '0': 0, 'Wake': 0,
    '1': 1, '2': 1, 'N1': 1, 'N2': 1,
    '3': 2, '4': 2, 'N3': 2,
    '5': 3, 'REM': 3, 'R': 3,
}
CLASS_NAMES = ['Wake', 'Light (N1+N2)', 'Deep (N3)', 'REM']
```

Figure 1. Label_Mapping

After extracting the ECG signal from the data, preprocessing was applied to improve the quality of the signal before feeding the signal into the model. The preprocessing steps included: Bandpass filtering, Notch filtering , Normalization.

Bandpass filtering was applied to the ECG signal to preserve only the useful frequency range related to cardiac activity while removing low-frequency baseline drift and high-frequency noise. This helped improve the quality of the ECG waveform before further analysis. Range: 0.5 Hz to 40 Hz because most important ECG information exists within this frequency range.

Notch filtering was used to remove power-line interference caused by electrical noise, typically at 60 Hz. This step was important because electrical interference can affect the accuracy of R-peak detection and HRV feature extraction.

Normalization was then applied to standardize the ECG signal amplitude across all epochs. This ensured that the model focused on learning meaningful waveform patterns rather than being affected by differences in signal scale between subjects.

After removing the noise and preprocessing, the Heart Rate Variability features are extracted from the clean signal. HRV feature extraction was performed to capture the physiological changes in heart activity during different sleep stages. First, R-peaks were detected from the ECG signal, then RR intervals were calculated as the time difference between consecutive heartbeats. Based on these intervals, several time-domain and frequency-domain HRV features were extracted, such as Mean RR, SDNN, RMSSD, Mean Heart Rate, and LF/HF ratio. These features help the model distinguish between different sleep stages more effectively.

The ECG signal was segmented into fixed intervals of 30 seconds. Each 30-second segment is referred to as an epoch, sampled at 256 Hz. So, Epoch length: $30s \times 256Hz = 7680$ samples. This segmentation made the signal easier to process and analyze. A 30-second epoch length was used because it is the standard duration commonly used in sleep staging. The model would consider a time window of several adjacent epochs. The context window consists of 11 consecutive epochs (5.5 minutes): five preceding epochs, the current epoch, and five following epochs. Sleep stages have a sequential nature, meaning that transitions between stages occur gradually rather than abruptly. So, the current stage is affected by the previous and the upcoming stages.

The data is then split into Training (7 subjects) , Validation (2 subject), Test (1subject). The data split was subject-wise rather than the epoch level to avoid data leakage. Dividing the data depending on the epoch means the same person is fed in the training and testing, which may cause the model to memorize subject-specific patterns instead of learning generalizable sleep patterns and lead to overfitting.

Total epochs: 12758

[0] Wake: 5569 (43.7%)

[1] Light (N1+N2): 5059 (39.7%)

[2] Deep (N3): 869 (6.8%)

[3] REM: 1261 (9.9%)

- Train: 8561 Epochs
- Val: 2,998 Epochs
- Test: 1,199 Epochs

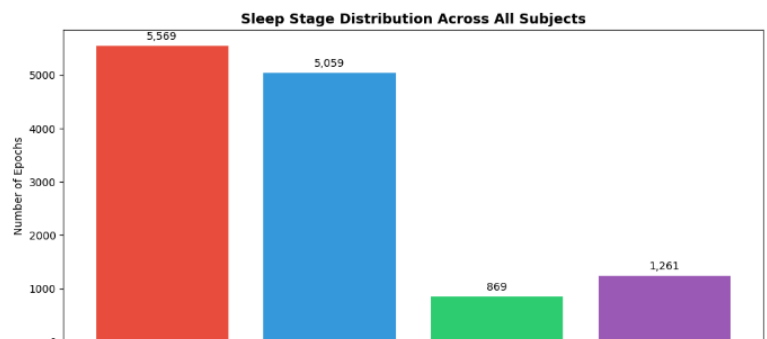


Figure 2. Class_Distribution

Challenges:

One of the main challenges in preprocessing was the class imbalance problem. Deep sleep and REM stages had significantly fewer samples compared to Wake and Light sleep stages. This imbalance biases the model toward the majority classes and reduces performance on minority classes. The limited number of subjects represented a major limitation. Deep learning models generally require large and diverse datasets to generalize effectively. Using only 10 subjects increased the risk of overfitting and reduced the model's ability to generalize to unseen subjects. The evaluation results confirmed the impact of class imbalance, as the model failed to correctly classify Deep Sleep, achieving an F1-score of 0.0 for this class. Most Deep were misclassified as Light sleep or REM sleep.

Another challenge was the instability of HRV features. Since HRV depends on accurate R-peak detection, noisy ECG segments produce unreliable RR intervals and noisy HRV measurements. This affects the quality of the extracted features.

8. Model Architecture

The model classifies sleep stages using both raw ECG waveforms and HRV features. The architecture combines CNN-based ECG feature extraction, BiLSTM temporal modeling, an attention mechanism, and HRV feature fusion to capture both ECG waveform patterns and temporal relationships between neighboring sleep epochs. Each input sample contains 11 consecutive epochs representing approximately 5.5 minutes of sleep data.

The waveform branch uses a CNN-based feature extractor called EpochCNN, which processes each ECG epoch independently to learn meaningful patterns from raw ECG signals.

The extracted ECG features are then passed into a 2-layer Bidirectional LSTM to learn temporal dependencies between neighboring sleep epochs. Since sleep stages transition gradually over time, using temporal context helps improve classification performance.

An attention mechanism is applied over the BiLSTM outputs to allow the model to focus on the most informative epochs within the context window.

In parallel, HRV features are processed through a separate projection branch and then fused with the temporal features extracted from the ECG waveform branch. The combined representation is finally passed through fully connected layers to predict the four sleep stages.

CNN layers were used because they are effective at learning local ECG waveform patterns directly from raw signals, while the BiLSTM captures sequential relationships between sleep stages. The attention mechanism improves the model by emphasizing the most relevant epochs during classification.

9. Model Training and Optimization

previously, we conducted a code with 72% accuracy but has overfitting. So, we will compare the current code with 70% accuracy but higher F1 and the previous code of 72%.

Project A: current project (70%)

```
Training configuration:
Loss          : FocalLoss (gamma=2.0)
Optimizer     : AdamW (lr=0.0003, weight_decay=2e-4)
Scheduler     : CosineAnnealingWarmRestarts
Epochs       : 30
Batch size    : 8
Grad clip     : 1.0
```

Figure 3. current_Training_Configuration

Project B: Previous project (72%)

```
Training configuration:
Loss          : CrossEntropyLoss (weighted, label_smoothing=0.1)
Optimizer     : AdamW (lr=0.0003, weight_decay=2e-4)
Scheduler     : Constant LR (no scheduling)
Epochs       : 30
Batch size    : 8
Grad clip     : 1.0
```

Figure 4. Training_Configuration_Previous

9.1 Loss Function and Class Imbalance Handling

Project_A:

The model was trained using Focal Loss with a gamma value of 2.0. Focal Loss is particularly effective for imbalanced classification tasks because it reduces the contribution of easily classified samples and forces the model to focus more on difficult and minority-class examples. This was important because the Deep Sleep and REM stages contained fewer samples compared to the Wake and Light Sleep stages.

To further address class imbalance, a Weighted Random Sampler was applied during training. The sampler increased the probability of selecting minority-class samples, helping the model learn balanced representations and reducing bias toward the majority classes.

Project_B:

The model was trained using a weighted CrossEntropyLoss with label smoothing. CrossEntropyLoss is appropriate for multi-class classification because the task involves predicting one of four sleep stages. Since the dataset is highly imbalanced, class weights were used to give more importance to the minority classes, especially Deep Sleep and REM, which appear much less frequently than Wake and Light Sleep. This helps reduce bias toward the majority classes and improves the model's ability to learn underrepresented sleep stages. label smoothing with a value of (0.1) was applied to reduce overconfidence in the predictions and improve generalization.

```
Class weights (to correct imbalance):  
[0] Wake: 0.542  
[1] Light (N1+N2): 0.616  
[2] Deep (N3): 5.012  
[3] REM: 3.002
```

Figure 5. Class_Weights

9.2 Optimizer and Regularization

The model was optimized using the AdamW optimizer with a learning rate of 3×10^{-4} . it was selected because it provides stable and efficient convergence for deep learning models while improving generalization through decoupled weight decay regularization.

A weight decay value of 2×10^{-4} was used to reduce overfitting and improve the model's ability to generalize to unseen data.

```
optimizer = optim.AdamW(  
    model.parameters(),  
    lr=LEARNING_RATE,  
    weight_decay=2e-4,  
    betas=(0.9, 0.999)  
)
```

Figure 6. Optimizer

The dropout rate was reduced compared to previous versions, and heavy label smoothing was removed. Excessive regularization was found to suppress important

discriminative ECG features, particularly for Deep Sleep and REM stages. Reducing these constraints helped the model learn more distinctive waveform patterns while maintaining stable generalization performance.

9.3 Data Augmentation and Mixup

Several augmentation techniques were applied to improve generalization. Random Gaussian noise, amplitude scaling, and temporal shifting were applied to ECG signals to increase variability and reduce overfitting.

Mixup augmentation was used during training using a mixing factor with $\alpha=0.2$. This technique helps the model learn smoother decision boundaries, improves generalization, and reduces sensitivity to noise and class memorization.

9.4 Gradient Accumulation and Gradient Clipping

Because of memory constraints, gradient accumulation was used to simulate a larger effective batch size. This makes training more stable and allows the model to benefit from a larger batch size even when GPU memory is limited.

gradient clipping was applied with a maximum norm of (1.0). This prevents exploding gradients and helps stabilize the training process.

9.5 Batch Size Configuration

The model was trained using a batch size of 8 with gradient accumulation steps of 4, resulting in an effective batch size of 32. This configuration allowed efficient training while balancing computational requirements and model performance.

9.6 Learning Rate Strategy

Project_A:

A Cosine Annealing Warm Restarts learning rate scheduler was used during training. The scheduler periodically adjusts the learning rate following a cosine decay pattern and restarts it after a fixed number of epochs to help the optimizer escape local minima and improve convergence.

The initial learning rate was set to 3×10^{-4} and the minimum learning rate was 1×10^{-6} . This adaptive learning rate strategy provided more stable optimization and improved training performance compared to using a constant learning rate.

Project_B:

We used a constant learning rate. This means the learning rate remained fixed at 3×10^{-4} throughout training.

10. Experimental Results

Since the dataset is imbalanced, F1-score was considered more informative than accuracy alone.

Part_A: The previous Project_B with 72%

10.1 Training curves

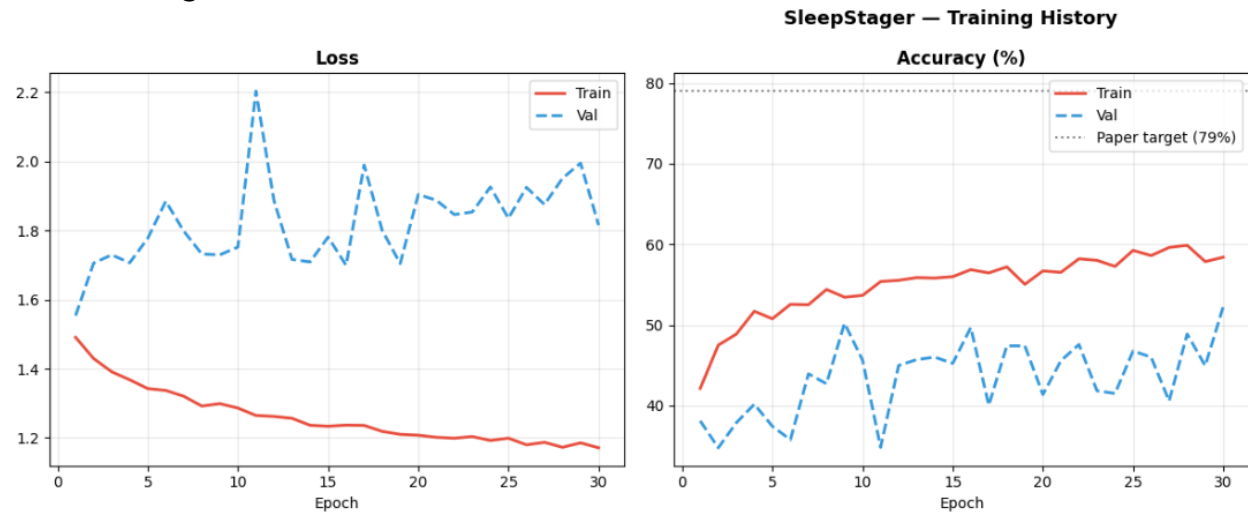


Figure 7. Training and validation loss and accuracy curves_Project_B

The training loss gradually decreased during training, which indicate that the model was progressively learning useful patterns from the ECG waveform and HRV features. The training loss curve was relatively decreasing smoothly suggesting stable optimization during training.

But the validation loss remained noticeably higher and fluctuated across training. This indicates that the model had difficulty generalizing to unseen data. suggesting the presence of overfitting.

The training accuracy gradually increase showing that the model successfully learned patterns from the training set. However, validation accuracy fluctuated roughly rather than increasing steadily which further reflects instability in the model's generalization performance.

The instability in validation performance is likely related to several factors. First, the dataset used in this project contained only 10 subjects, which is relatively small for training a deep learning architecture with CNN, BiLSTM, and attention components. Second, the dataset suffered from severe class imbalance, particularly for Deep Sleep and REM stages, which made generalization more difficult. ECG signals naturally vary between subjects, which may further increase variability in validation performance.

The gradual improvement in training accuracy demonstrates that the architecture was capable of extracting useful physiological and temporal sleep-related features from ECG signals. The use of Mixup augmentation, weighted loss, and regularization techniques helped stabilize training and reduce more severe overfitting.

The final test accuracy was higher than the validation accuracy observed during training because the final evaluation used the best saved checkpoint together with Test-Time Augmentation (TTA), which improved prediction robustness. In addition, the small number of validation subjects may have contributed to higher variability in validation performance.

10.2 Confusion matrix

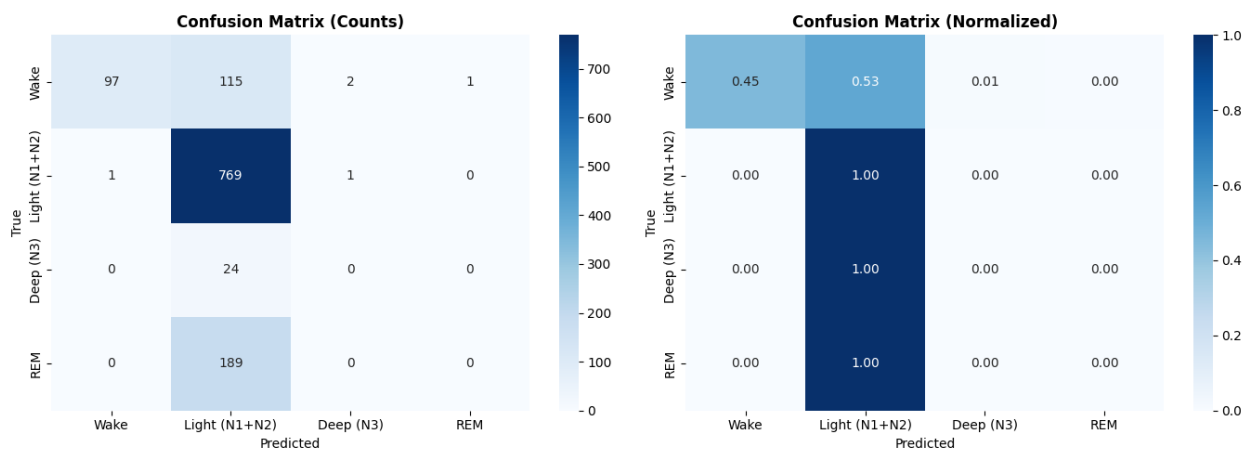


Figure 8. Confusion_Matrix_Project_B

It shows strong performance in classifying the Light Sleep stage. Most Light Sleep samples were correctly classified, indicating that the model learned features associated with this stage more effectively than the other stages. This is likely because Light Sleep represented a large portion of the dataset, giving the model more training examples for this class.

However, the model struggled significantly with Deep Sleep and REM classification. Most Deep Sleep and REM samples were incorrectly predicted. The model only classified 3 samples as deep and 1 sample as REM, this suggests that the model was biased against predicting these minority classes as the final output. even though the model still assigned probabilities to these classes during prediction. Deep and REM are totally classified as light. Therefore, the model was relatively better at distinguishing Wake from Light Sleep but struggled to separate Light Sleep from the other sleep stages.

The Wake stage achieved moderate performance. Some Wake samples were correctly classified, while many others were confused with Light Sleep. This confusion

may occur because transitions between Wake and Light Sleep can contain similar physiological patterns in ECG and HRV signals.

The confusion matrix also reflects the strong class imbalance present in the dataset. Since Light Sleep had a much larger number of training samples compared to Deep Sleep and REM, the model became biased toward predicting the majority class. As a result, the classifier tended to favor Light Sleep predictions when uncertainty was present.

Another possible reason for the poor Deep Sleep and REM performance is that ECG-only signals may not contain enough discriminative information to clearly separate all sleep stages. Stages such as Deep Sleep and REM are often more accurately identified using additional physiological signals such as EEG or EOG.

Despite these limitations, the confusion matrix confirms that the model was still able to learn meaningful sleep-related patterns from ECG signals, particularly for distinguishing Light Sleep from Wake states. With further improvements, the model could potentially be used in driver drowsiness detection.

10.3 ROC curves

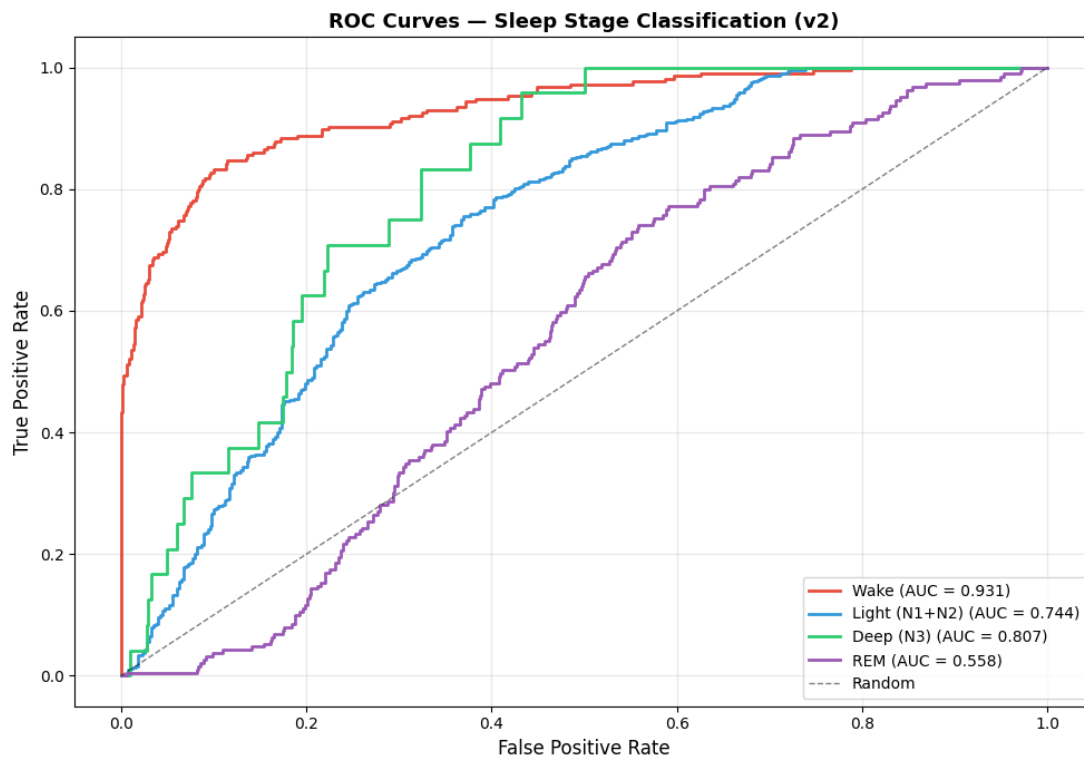


Figure 9.ROC_CURVE_Project_B

The ROC curves show that the model achieved the best performance for the Wake stage with an AUC of 0.931, indicating strong separability between Wake and the other

sleep stages. Deep Sleep also showed relatively good performance with an AUC of 0.807, while Light Sleep achieved moderate performance with an AUC of 0.744.

Although the confusion matrix showed poor final classification for Deep Sleep, the ROC curve suggests that the model was still able to assign meaningful probabilities to this class internally. This means the model partially learned Deep Sleep characteristics but struggled when making the final classification decision.

In contrast, REM classification showed weak performance with an AUC of 0.558, which is close to random classification. This suggests that the model had difficulty learning distinctive ECG and HRV patterns for REM sleep. The lower REM performance is likely related to class imbalance, limited training data, and overlap between physiological characteristics of REM and other sleep stages.

10.4 Classification report

Classification Report:				
	precision	recall	f1-score	support
Wake	0.9898	0.4512	0.6198	215
Light (N1+N2)	0.7010	0.9974	0.8233	771
Deep (N3)	0.0000	0.0000	0.0000	24
REM	0.0000	0.0000	0.0000	189
accuracy			0.7223	1199
macro avg	0.4227	0.3621	0.3608	1199
weighted avg	0.6283	0.7223	0.6406	1199

Figure 10. Classification_Report_Project_B

FINAL PERFORMANCE SUMMARY – SleepStager				
	Class Precision (%)	Recall (%)	F1-score (%)	Support
Wake	98.98	45.12	61.98	215
Light (N1+N2)	70.10	99.74	82.33	771
Deep (N3)	0.00	0.00	0.00	24
REM	0.00	0.00	0.00	189

Overall Accuracy : 72.23%				
Macro F1-score : 36.08%				
Macro AUC : 0.7599				
=====				
Reference paper : ~79% accuracy (Zhang et al. 2023)				
=====				

Figure 11. Final_Summary_Project_B

The final model achieved an overall accuracy of 72.23%, which is relatively close to the reference paper performance, considering that this project used only 10 subjects compared to more than 1000 subjects in the paper. The Macro AUC reached 0.7599,

showing that the model was still able to learn useful class separability from ECG and HRV signals.

The best performance was achieved for the Light Sleep stage, with a recall of 99.74% and an F1-score of 82.33%. This indicates that the model was highly biased toward predicting the Light Sleep class, which is expected due to the class imbalance in the dataset.

Wake classification achieved high precision (98.98%), meaning that when the model predicted Wake, it was usually correct. However, the recall for Wake was much lower (45.12%), showing that many Wake samples were still confused with Light Sleep.

Deep Sleep and REM stages achieved very poor performance, with precision, recall, and F1-score values equal to 0. This indicates that the model failed to correctly classify these stages in the final predictions, even though the ROC analysis showed that the model still learned partial probability representations for these classes internally.

The low Macro F1-score (36.08%) further highlights the imbalance in class performance. While the model performed well on majority classes, it struggled significantly with minority stages such as Deep Sleep and REM.

10.5 Hypnogram comparison

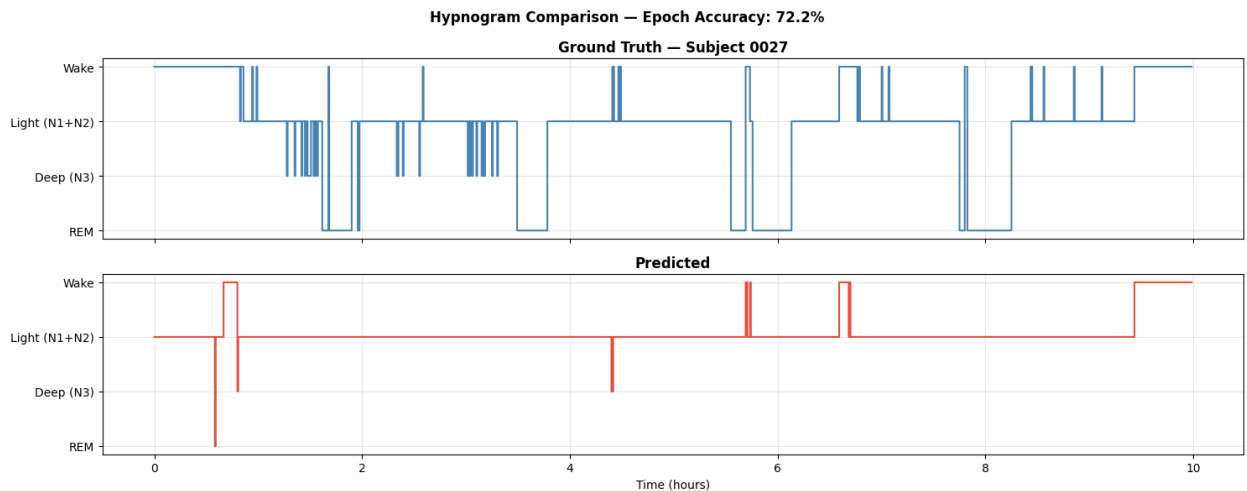


Figure 12. Hyponogram_Project_B

The hypnogram comparison shows the difference between the ground truth sleep stages and the model predictions over time for Subject 0027. The predicted hypnogram generally followed the overall sleep structure and achieved an accuracy of 72.2%.

The model was able to identify Wake and Light Sleep stages reasonably well, but it struggled to detect Deep Sleep and REM stages consistently. Most Deep Sleep and REM

epochs were predicted as Light Sleep, which is consistent with the confusion matrix and classification report results.

This behavior highlights the effect of class imbalance and the difficulty of distinguishing minority sleep stages using ECG and HRV signals alone.

Part_B: Project_A with 70%

10.6 Training curves :

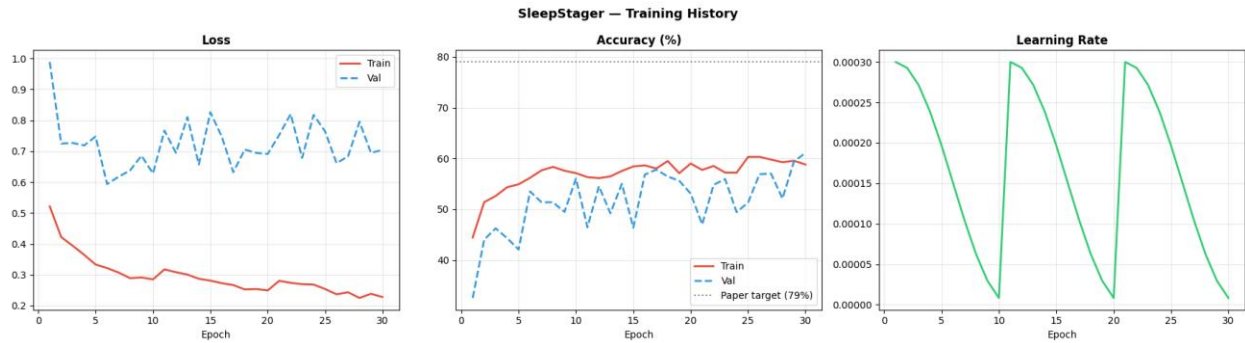


Figure 13. Training curves Project_A

The training loss decreased steadily throughout training, while the validation loss fluctuated at a higher range. This indicates that the model successfully learned meaningful ECG representations, but still faced difficulties generalizing consistently to unseen subjects. The instability in validation loss is mainly associated with the limited dataset size and severe class imbalance, particularly for Deep Sleep and REM stages.

Training and validation accuracy both improved gradually during training, with validation accuracy reaching nearly 60% by the final epochs. The relatively small gap between training and validation accuracy suggests that the applied regularization techniques, including Mixup augmentation, weight decay, Focal Loss, and gradient clipping, helped reduce severe overfitting compared to the previous model configuration. Although the achieved accuracy remained below the paper target, the model demonstrated more stable generalization behavior.

The learning rate curve reflects the behavior of the Cosine Annealing Warm Restarts scheduler. The periodic reduction and restart of the learning rate allowed the optimizer to avoid early convergence and improved stability during training. This indicates that adaptive learning rate scheduling contributed positively to the convergence behavior of the model.

10.7 Confusion matrix

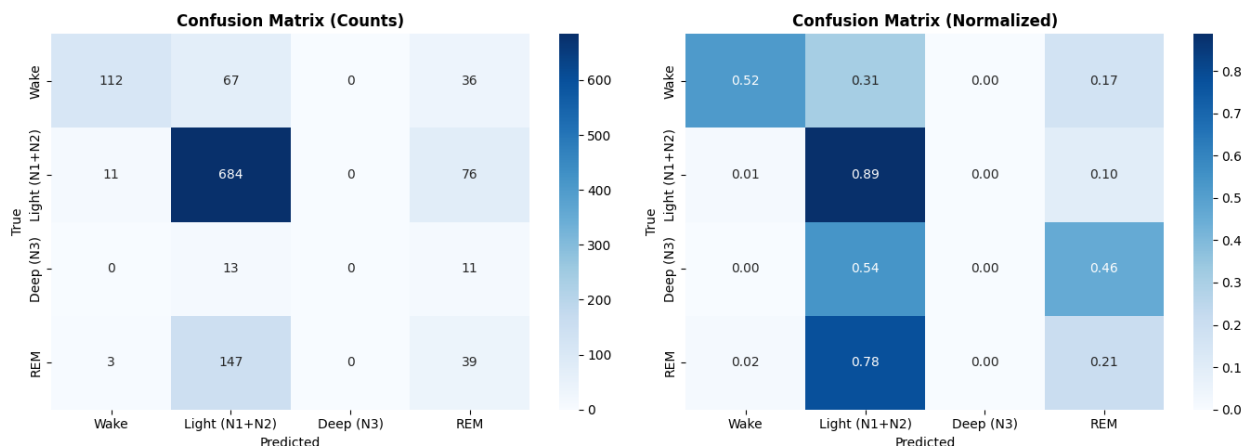


Figure 14. Confusion_Matrix_Project_A

The model achieved strongest performance on the Light Sleep stage, correctly classifying approximately 89% of Light epochs. This indicates that the extracted ECG and HRV features were effective in capturing the dominant patterns associated with light sleep.

Wake stages were classified with moderate performance, achieving about 52% correct predictions. A noticeable portion of Wake epochs was misclassified as Light sleep, suggesting that transitions between these stages share similar physiological characteristics in ECG signals.

The model showed poor performance on Deep Sleep. None of the Deep Sleep epochs were correctly classified, with most samples misclassified as Light sleep or REM sleep. This indicates that the model struggled to learn discriminative features for Deep Sleep. Also it is noticed that the model didn't choose deep stage as final decision for any sample. Zero classified as deep.

REM sleep also remained challenging, with only 21% of REM epochs classified correctly. Most REM samples were predicted as Light sleep, indicating that the ECG patterns of REM and Light stages overlapped significantly. Unlike the previous model, in this model, it classified more samples as REM. The model is able to choose REM as the final decision.

10.8 ROC curves

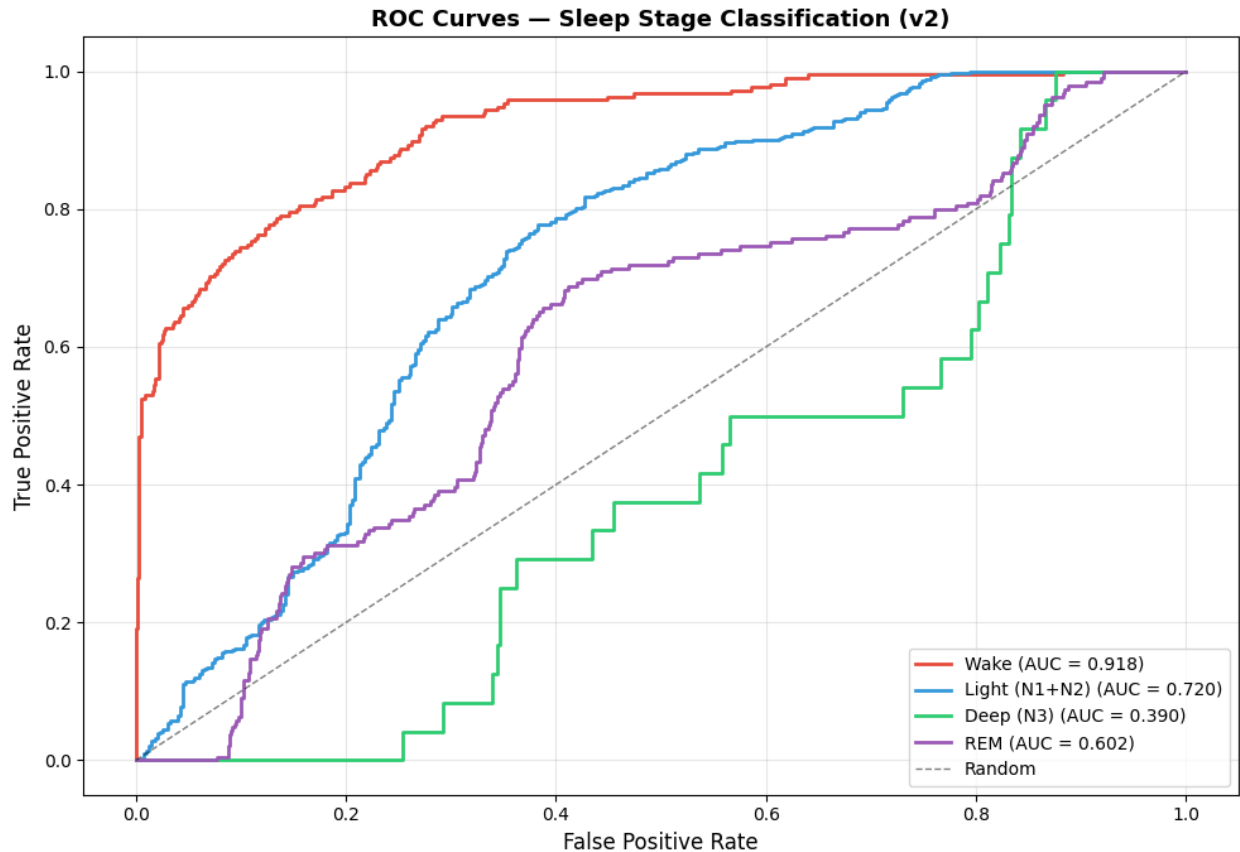


Figure 15.ROC_Curve_Project_A

The model achieved the highest performance for the Wake stage with an AUC of 0.918, indicating strong discriminative capability between Wake and the other sleep stages. This suggests that Wake-related ECG patterns were more distinct and easier for the model to learn.

The Light Sleep stage achieved an AUC of 0.720, reflecting moderate classification performance. Since Light sleep represents the majority class in the dataset, the model learned its characteristics more effectively compared to minority classes.

Performance for REM sleep was lower, with an AUC of 0.602, indicating limited separability between REM and other stages. The overlap between REM and Light sleep ECG patterns likely contributed to this reduced performance.

The weakest performance was observed for Deep Sleep, with an AUC of 0.390. This value is below random classification performance, demonstrating that the model failed to learn reliable discriminative features for Deep Sleep.

10.9 Classification report

Classification Report:				
	precision	recall	f1-score	support
Wake	0.8889	0.5209	0.6569	215
Light (N1+N2)	0.7508	0.8872	0.8133	771
Deep (N3)	0.0000	0.0000	0.0000	24
REM	0.2407	0.2063	0.2222	189
accuracy			0.6964	1199
macro avg	0.4701	0.4036	0.4231	1199
weighted avg	0.6801	0.6964	0.6758	1199

Figure 16. *Classificatio_Report_Project_A*

=====				
FINAL PERFORMANCE SUMMARY – SleepStager				
=====				
Class	Precision (%)	Recall (%)	F1-score (%)	Support
Wake	88.89	52.09	65.69	215
Light (N1+N2)	75.08	88.72	81.33	771
Deep (N3)	0.00	0.00	0.00	24
REM	24.07	20.63	22.22	189

Overall Accuracy : 69.64%				
Macro F1-score : 42.31%				
Macro AUC : 0.6576				
=====				
Reference paper : ~79% accuracy (Zhang et al. 2023)				
=====				

Figure 17. *Final_Summary_Project_A*

The classification report shows that the model achieved an overall accuracy of 69.64% and a macro F1-score of 42.31%. The results indicate that the model performed well on majority classes but struggled significantly with minority sleep stages.

Light Sleep achieved the best performance with an F1-score of 81.33% and recall of 88.72%, showing that the model learned Light sleep patterns effectively. Wake also achieved relatively strong precision (88.89%), indicating reliable Wake predictions when detected correctly.

In contrast, REM classification performance remained limited, with an F1-score of only 22.22%. Most REM epochs were misclassified as Light sleep, suggesting overlapping ECG characteristics between these stages.

The weakest performance was observed for Deep Sleep, where the model failed to correctly classify any samples, resulting in an F1-score of 0.0%. This strongly reflects the

impact of severe class imbalance, as Deep Sleep represented only a small portion of the dataset.

10.10 Hypnogram comparison

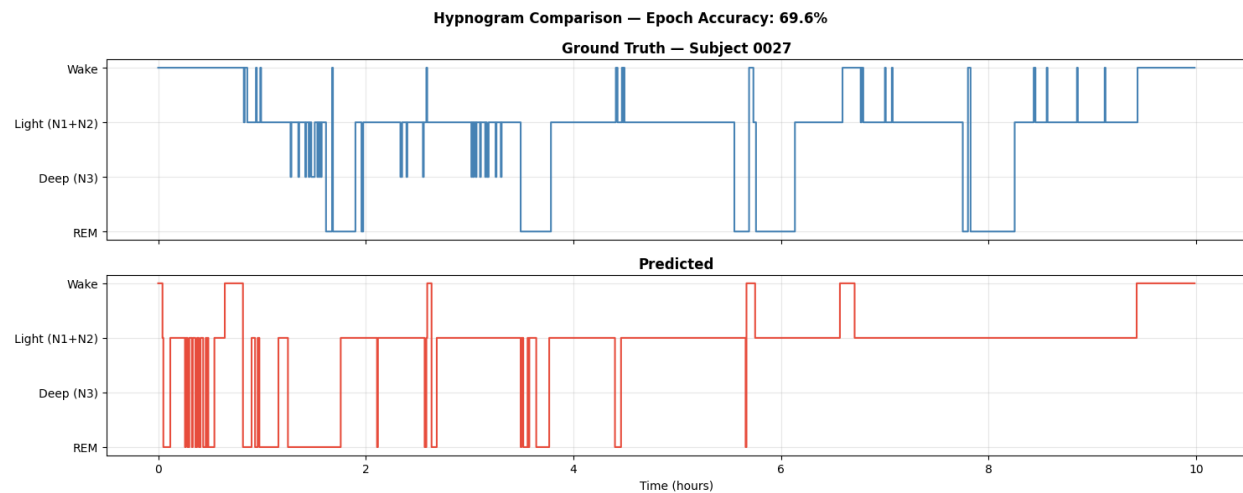


Figure 18. Hypnogram_Comparison_Project_A

The model successfully captured the overall sleep structure and major transitions between Wake and Light sleep stages, which contributed to the achieved accuracy of 69.6%.

Most prediction errors occurred during Deep Sleep and REM stages. The predicted hypnogram shows that the model tended to classify many epochs as Light sleep, even when the true stage was Deep Sleep or REM. This behavior indicates that the model was biased toward the majority Light Sleep class because of dataset imbalance.

Despite these limitations, the predicted hypnogram still followed the general temporal progression of sleep stages, suggesting that model were able to learn meaningful sequential sleep patterns from ECG signals.

11. Comparison Between Project_A (Current model) and Project_B (Previous model)

Metric	Project A	Project B
Accuracy	69.64%	72.23%
Macro F1 score	42.31%	36.08%
Macro AUC	0.6576	0.7599
Wake AUC	0.918	0.931
Light Sleep AUC	0.720	0.744
Deep Sleep AUC	0.390	0.807
REM AUC	0.602	0.558
Wake F1	65.69%	61.98%
Light Sleep F1	81.33%	82.33%
Deep Sleep F1	0.00%	0.00%
REM F1	22.22%	0.00%
Overfitting	Lower	Higher
Generalization	Better	poor

Project B achieved a slightly higher overall accuracy of 72.23% compared to 69.64% for Project A. However, the higher accuracy of Project B was mainly caused by strong bias toward the majority Light Sleep class. The classification report and confusion matrix showed that Project B completely failed to classify Deep Sleep and REM stages, with F1 scores equal to 0 for both classes. This indicates severe overfitting and poor generalization to minority sleep stages.

In contrast, Project A achieved lower overall accuracy but demonstrated more stable and balanced learning behavior. The use of Focal Loss, Cosine Annealing Warm Restarts reduced overfitting and improved validation stability. Unlike Project B, Project A was able to classify some REM samples correctly and achieved a higher Macro F1 score (42.31% vs 36.08%), indicating better overall class balance.

The training curves of Project A also showed a smaller gap between training and validation accuracy, suggesting improved generalization performance. Although Deep Sleep classification remained unsuccessful in both models, Project A produced more reliable probability distributions and less biased predictions.

Although Project B achieved a higher Macro AUC, this does not necessarily indicate better final classification performance. AUC measures the model's ability to separate classes probabilistically across different decision thresholds, while F1 score evaluates the quality of the final class predictions by balancing precision and recall. In Project B, the model was still able to assign meaningful probabilities to minority classes such as Deep

Sleep internally, which resulted in a relatively high AUC. However, during the final classification decision, the model became strongly biased toward the majority Light Sleep class, causing Deep Sleep and REM F1 scores to drop to 0. In contrast, Project A achieved better Macro F1 score and more balanced predictions, indicating improved generalization and reduced class bias despite the lower overall AUC.

Therefore, Project A can be considered the better model overall because it achieved more balanced classification performance and reduced overfitting, even with slightly lower accuracy. Since the dataset is highly imbalanced, Macro F1-score and generalization behavior are more informative evaluation measures than accuracy alone. In our case the accuracy is a misleading value as the data is unbalanced.

12. Future Work

One important direction is increasing the dataset size and using a larger number of subjects to improve model generalization and reduce overfitting. The severe class imbalance problem, particularly for Deep Sleep and REM stages, can also be addressed using more advanced balancing techniques.

Multimodal physiological signals such as EEG, EOG or respiration signals could improve the classification of difficult sleep stages that are not easily distinguishable using ECG alone.

Try to separate the model into 2 parts. The first part classifies Wake, Light and Other. the second part classifies the Other into deep sleep or REM.

Cross dataset evaluation and testing on larger subject independent datasets may also help assess the robustness and generalization capability of the model.

The Project could be further improved and be integrated into driver drowsiness detection applications for real-time sleep.

13. Conclusion

To conclude, this project developed a deep learning model for automated sleep stage classification using only ECG channels from the MESA Sleep Study dataset. The model uses a Convolutional Neural Network for per epoch waveform feature extraction, a bidirectional LSTM for capturing sleep dynamics, and a Heart Rate Variability fusion branch. Due to the obvious class imbalance, several training stages were used throughout the pipeline, like Focal Loss, WeightedRandomSampler, Mixup augmentation, and Cosine Annealing Warm Restarts scheduling. The final model was tested on a held out test using Test-Time Augmentation. It achieved an overall accuracy of 69.64% and a macro F1 score of 42.31%.

Throughout model development process from the beginning of the project, several optimization and imbalance-handling techniques were explored, including Weighted CrossEntropy Loss, Balanced Softmax, Focal Loss, Label Smoothing, WeightedRandomSampler, Mixup augmentation, ECG augmentation techniques, Cosine Annealing Warm Restarts scheduling, Test-Time Augmentation (TTA), Gradient Clipping, Gradient Accumulation, and Dropout tuning.

Despite relatively good overall accuracy, the low macro F1 score highlights the challenge we faced: good performance on dominant classes (light sleep) but significant struggles with minorities (deep sleep and REM). This gap in scores shows us that accuracy alone is not enough to evaluate the performance of a model. In addition to that, the small number of samples used, 10 subjects, in this project was another limitation to training and testing. Increasing the number of subjects would help the model train more and thus give better and more reliable results.

In terms of future work, it could address the class imbalance in a better way. Incorporating additional physiological signals could also meaningfully improve minority class recall and improve overall performance.

14. References

C. Won et al., “Sex differences in obstructive sleep apnea phenotypes: The multi-ethnic study of atherosclerosis,” *Sleep*, vol. 43, no. 1, p. zsz274, 2019.

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(<https://my.clevelandclinic.org/health/body/12148-sleep-basics>)

The AASM Manual for the Scoring of Sleep and Associated Events.

(<https://fullpdfword.com/look-up/mL6C54/603299/5020536-aasm-manual-for-the-scoring-of-sleep>)

J. Huo, H. Li, S. F. Quan, J. Roveda, and A. Li, “A Multi-task Deep Learning Algorithm for Sleep Stage Scoring and Sleep Arousal Detection,” *Frontiers in Sleep*, 2023.

15. Appendix

Previous Versions Results (There are more versions but those are sample of the progress):

V1:

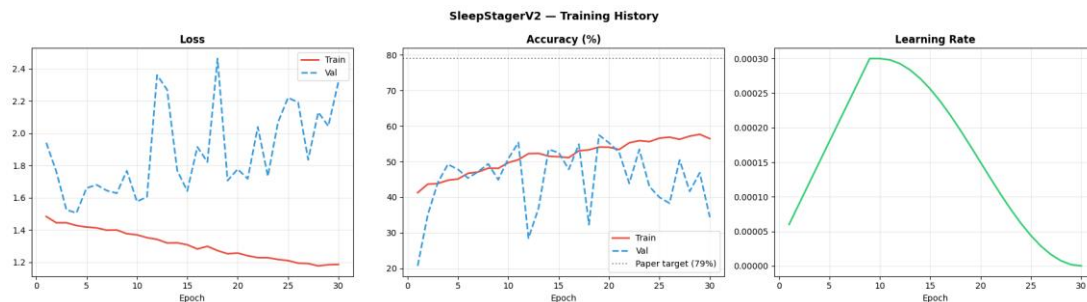


Figure 19. V1_TrainingCurves

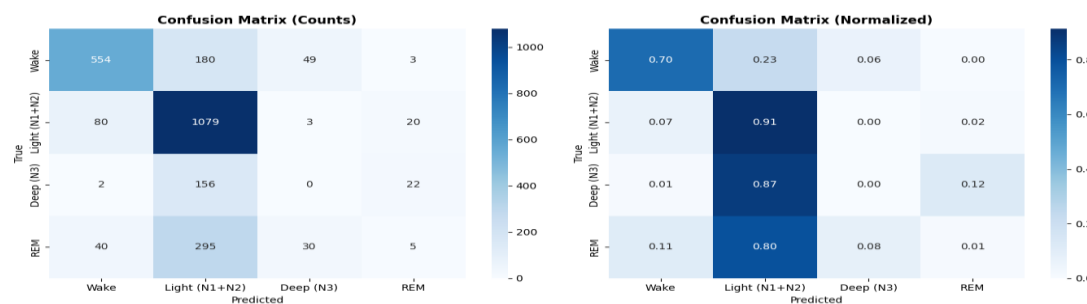


Figure 20. V1_ConfusionMatrix

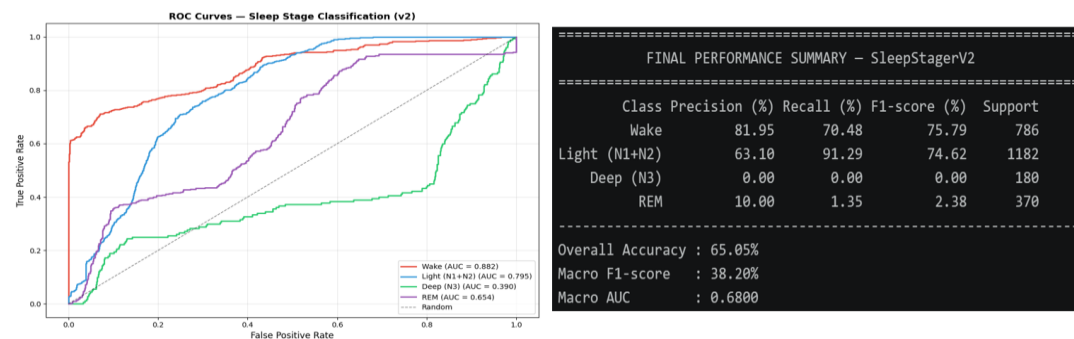


Figure 21. V1_ROC&FinalSummary

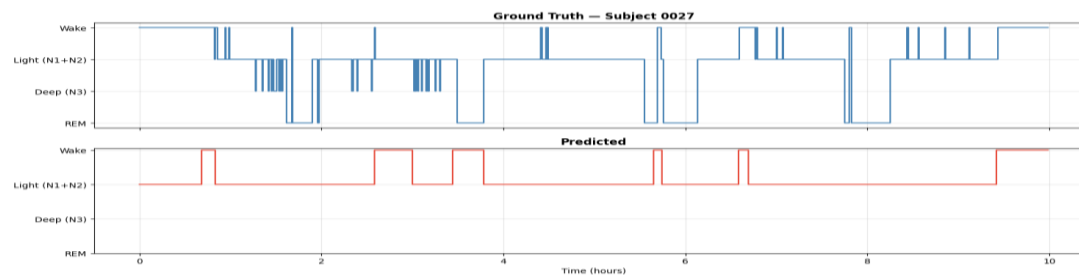


Figure 22. V1_Hyponogram

V2:

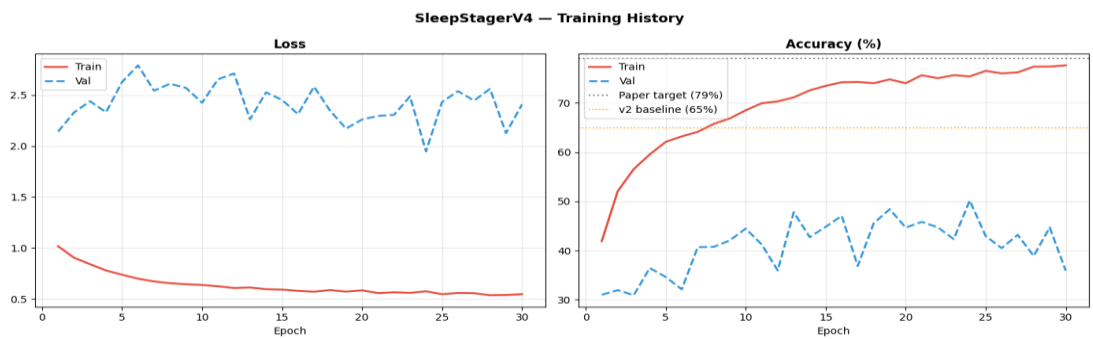


Figure 23. V2_TrainingCurves

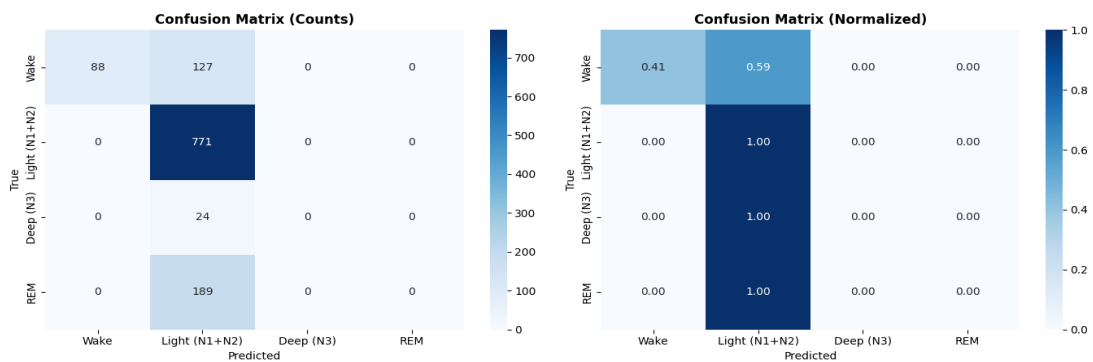


Figure 24.V2_ConfusionMatrix

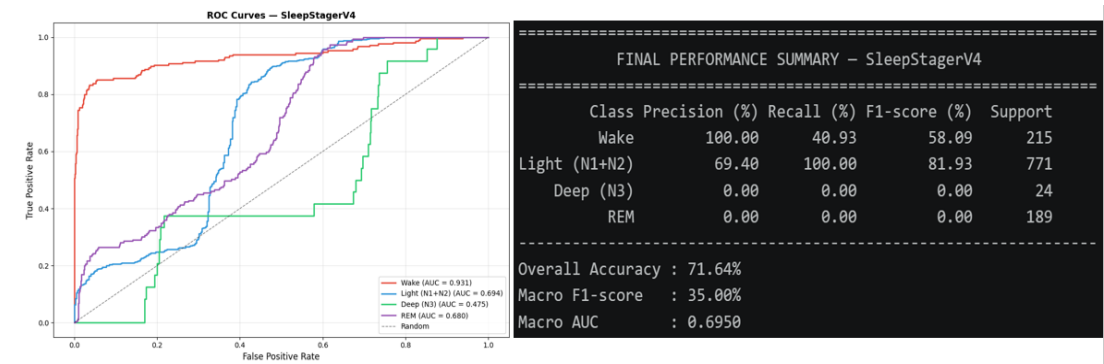


Figure 25.V2_ROC&FinalSummary

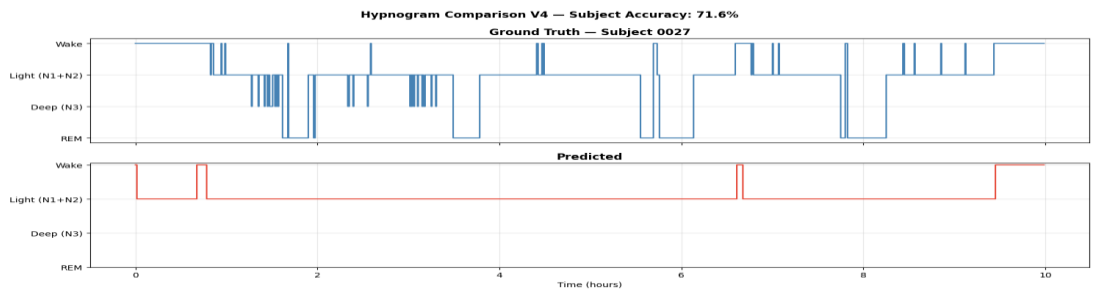


Figure 26 V2_Hyponogram

V3:

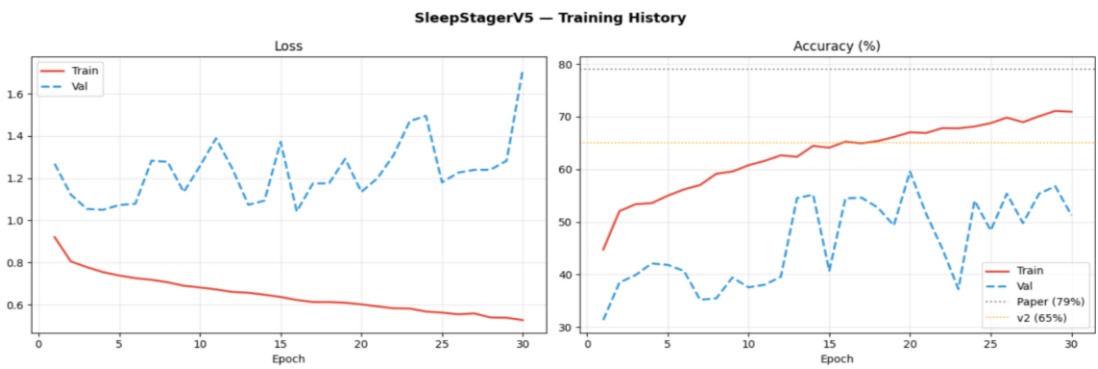
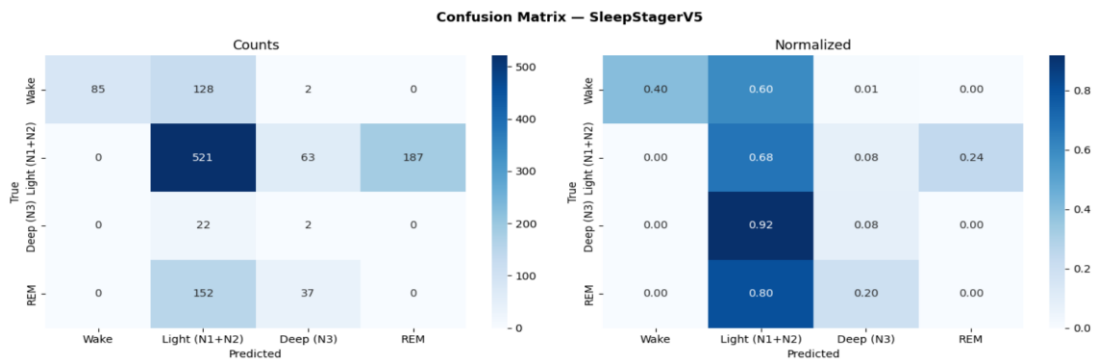


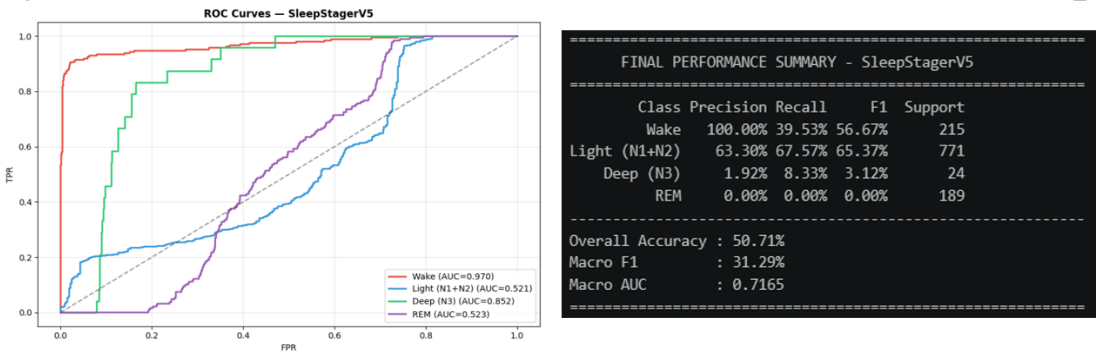
Figure 27 V3_TrainingCurves



Figure

28

V3_ConfusionMatrix



Figure

29

V3_ROC&FinalSummary

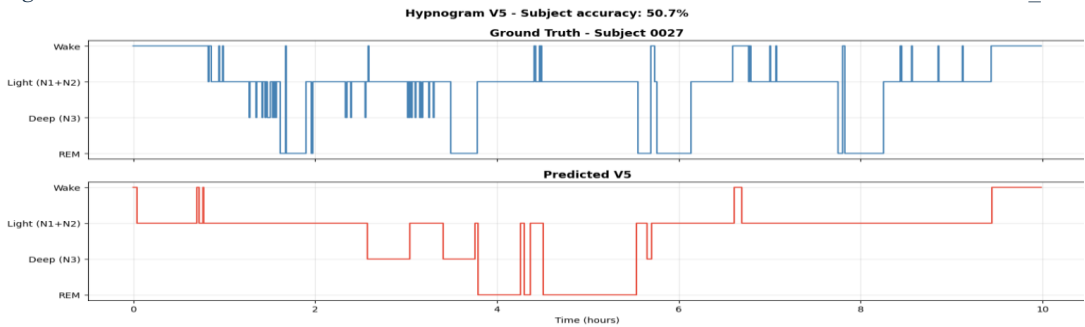


Figure 30 V3_Hypnogram